

Final project option 1: Survey on AI in Network Security

Objective: To conduct a rigorous, critical, and insightful survey of a specific application area where Artificial Intelligence (AI) and Machine Learning (ML) are being applied to network security. This project aims to develop your skills in academic research, critical analysis, technical writing, and the articulation of future research directions.

1. Project Overview

You will select a specific research topic within the domain of "AI for Network Security," conduct a comprehensive literature review, and write a survey paper in the style of a high-quality ACM or IEEE journal survey.

The goal is not simply to summarize papers, but to:

- Synthesize the state-of-the-art.
- Categorize existing approaches based on their methodology, application, and underlying AI/ML techniques.
- Critically evaluate the strengths, weaknesses, and assumptions of these approaches.
- Identify open challenges, research gaps, and emerging trends.
- Propose potential future directions for the field.

2. Learning Outcomes

Upon completion of this project, you will be able to:

- Formulate a focused research question within a broad domain.
- Navigate and utilize academic research databases (e.g., IEEE Xplore, ACM Digital Library, arXiv, Scopus).
- Critically read, analyze, and synthesize technical research papers.
- Construct a coherent taxonomy to organize a body of research.
- Evaluate the practical implications, limitations, and security assumptions of AI-driven network security systems (e.g., adversarial ML, data poisoning, model drift).
- Communicate complex technical concepts in a structured and professional written format.

3. Topic Selection

You are free to choose any topic where AI/ML is applied to a problem in network security. Topics should be specific enough to allow for deep analysis (e.g., "Intrusion Detection Systems") but broad enough to have a substantial body of literature (at least 15-20 core papers).

Choose one from the list below, or propose your own topic (subject to instructor approval by **Monday April 13**).

- AI-Powered Network Intrusion Detection Systems (NIDS)
- Malicious Traffic Analysis
- Phishing & Social Engineering Detection
- IoT/IIoT Network Security
- Adversarial Machine Learning in Network Security
- AI for Security Orchestration, Automation, and Response (SOAR)
- User and Entity Behavior Analytics (UEBA)
- AI-Assisted Threat Hunting
- Natural Language Processing (NLP) for Threat Intelligence
- AI in Zero Trust Architecture (ZTA).

- Federated Learning for Privacy-Preserving Network Security
- AI for cloud misconfiguration detection
- ML/DL for anomaly detection in network traffic
- AI (LLMs) in Data poisoning in network datasets
- AI-assisted vulnerability discovery
- AI (LLMs) in Prompt injection & misuse
- AI-generated phishing, deepfakes, malware
- Automated reconnaissance and lateral movement
- New “network-of-agents” threat models

4. Project Structure & Deliverables

The final deliverable is a survey paper (≥ 8 pages, single-column, 1.5line-spaced, 12pt font) . It must adhere to the following structure:

1. **Title and Abstract:** A concise (250-300 words) summary of the survey's scope, main categorization, key findings, and identified research gaps.
2. **Introduction:** Briefly describe the importance of network security, the challenges traditional methods face, and the motivation for applying AI. State the scope, contributions, and organization of your paper.
3. **Background:** Briefly describe relevant background for readers unfamiliar with the topic. *Do not assume the reader is an AI expert.*
4. **Taxonomy & Categorization:** This is the core of your survey. Propose a novel or refined taxonomy to organize the literature. This could be based on:
 - AI/ML Technique: (e.g., CNNs, RNNs, GANs, Reinforcement Learning).
 - Network Layer or Data Source: (e.g., packet-based, flow-based, log-based, host-based).
 - Security Goal: (e.g., detection, prevention, response, prediction).
 - Deployment Architecture: (e.g., cloud-based, edge-based, distributed).

For each category in your taxonomy, summarize the key approaches, papers, and recent advances.

5. **Critical Analysis:** Go beyond summarization. Critically evaluate the state of the art. Discuss:
 - **Datasets:** What datasets are commonly used? Are they realistic, representative, and up-to-date (e.g., CIC-IDS, UNSW-NB15, real-world PCAPs)? What are their limitations?
 - **Evaluation Metrics:** Are papers using appropriate metrics (e.g., precision, recall, F1-score, FAR, AUC)? Is there a focus on real-world performance (e.g., latency, throughput, false positive rate)?
 - **Limitations:** Discuss common weaknesses in the literature (e.g., lack of robustness to adversarial attacks, concept drift, explainability, computational cost, privacy implications).
 - **Adversarial ML:** Explicitly consider the security of the AI models themselves.
6. **Open Challenges & Future Research Directions:** Propose a list of 2-5 concrete, well-justified open problems and promising avenues for future research.
7. **Conclusion:** A brief summary of the survey's main findings and concluding remarks.
8. **References:** A comprehensive bibliography of the papers you cited, formatted in IEEE or ACM style. Aim for a minimum of 20 high-quality references from reputable conferences or journals.

5. Grading Rubric (Total: 100 points)

- **Topic Selection & Scope (10 pts):** Topic is appropriately focused, timely, and relevant. Scope is well-defined and manageable.
- **Literature Coverage & Depth (25 pts):** Demonstrates a comprehensive understanding of the field. Covers seminal papers and recent advances. Bibliography is robust and of high quality. Goes beyond the obvious papers.
- **Taxonomy & Synthesis (20 pts):** Presents a clear, logical, and insightful taxonomy for categorizing literature. Effectively synthesizes findings across multiple papers rather than listing them.
- **Critical Analysis & Evaluation (20 pts):** Provides a deep, insightful critique of the field. Identifies limitations, biases, and weaknesses in existing work. Discusses datasets, metrics, and assumptions effectively. Considers the security of the AI/ML models themselves.
- **Future Directions (10 pts):** Proposes insightful, concrete, and well-justified future research directions that stem logically from the analysis.
- **Writing Quality & Organization (10 pts):** Paper is well-structured, clear, and concise. Writing is professional, grammatically correct, and follows the required format. Figures and tables are used effectively to enhance clarity.
- **References & Formatting (5 pts):** References are complete, correctly formatted (IEEE/ACM), and consistently cited within the text. Paper adheres to length and formatting guidelines.

6. Academic Integrity

You can work individually or with another student as a group (one submission per group). All sources must be properly cited. Plagiarism, including reusing text from source papers without attribution, will result in a grade of zero. The use of AI text generators (e.g., ChatGPT) for writing the paper is prohibited. These tools may be used for brainstorming or finding keywords but not for generating substantive content.

Final project option 2: Research paper on Network Security

Objective: To conduct original research by identifying a gap or limitation in an existing network security solution, proposing an improvement, implementing it, and rigorously evaluating its performance. This project aims to develop your skills in conducting an original research study in network security and produce an academic-style paper.

1. Project Overview

You will build upon the knowledge gained from course lectures, readings, and research presentations to conduct a substantive research project. You will identify a specific problem in network security, review relevant literature, propose a novel improvement or extension to an existing approach, implement your solution, and evaluate its performance through systematic experimentation. The goal is to produce a complete research paper that could serve as a foundation for a conference submission.

Each student (or group) will:

- Identify a specific research problem in network security
- Study relevant prior work
- Design and implement a solution, improvement, or experimental study
- Evaluate results and compare them with existing approaches

2. Project Expectations

Your work must include at least one of the following:

- **Implementation + Improvement (Preferred)**
 - Reproduce an existing method (e.g., IDS, anomaly detection model)
 - Propose a **clear improvement**, such as:
 - Better accuracy
 - Lower false positives
 - Reduced computational cost
 - Improved robustness (e.g., adversarial resistance)
- **Experimental Comparative Study**
 - Implement or use multiple techniques
 - Perform controlled experiments
 - Provide insightful comparisons and analysis
- **System Design / Prototype**
 - Build a working security system or tool, such as:
 - Network monitoring tool
 - AI-based intrusion detection system
 - Traffic classification pipeline
 - Evaluate its effectiveness

3. Deliverables

The final deliverable is a research paper (≥ 5 pages, single-column, 1.5line-spaced, 12pt font) . It should adhere to the following structure:

- **Title and Abstract**
- **Introduction**
 - Set the context: the broader network security problem.

- Identify the specific gap or limitation in existing work.
 - State your proposed improvement and its key ideas.
 - Summarize your contributions
 - Outline the paper structure
- **Background and Related Work**
 - Provide necessary background for readers
 - Survey related work: what has been done, what are the limitations, and how your work differs and advances the state of the art.
 - Clearly position your contribution within the existing literature.
- **Problem Statement and Threat Model**
 - Precisely define the problem you are solving.
 - Explicitly state your assumptions and threat model: what capabilities does the attacker have? What are the system's trust assumptions? etc.
- **Proposed Approach**
 - Describe your method in detail. This section should be sufficiently clear that a knowledgeable researcher could reproduce your work.
 - Include: system architecture diagram, algorithm descriptions, mathematical formulations, design decisions and justifications.
 - If improving an existing method, clearly describe both the baseline and your novel contribution.
- **Implementation**
 - Describe your implementation details: programming language, frameworks/libraries (e.g., PyTorch, TensorFlow, Scapy, P4), hardware/software environment.
 - Discuss any non-trivial implementation challenges and how you addressed them.
- **Experimental Evaluation**
 - Research Questions: Explicit questions your experiments aim to answer (e.g., RQ1: Does our improvement reduce false positive rate compared to baseline?)
 - Experimental Setup
 - Datasets
 - Baselines: Clear description of comparison methods
 - Metrics: Performance metrics
 - Results and Analysis: Interpret what the results mean. Discuss why certain results occurred. Acknowledge limitations and unexpected findings.
- **Discussion**
 - Broader implications of your work, including limitations and Future work directions
- **Conclusion**
 - Brief summary of the problem, your contribution, and key findings.
- **References**
 - Complete bibliography in IEEE or ACM format. Minimum 15-20 high-quality references.
- **Appendix (Optional)**
 - Detailed parameter settings, full experimental results, or code snippets that don't fit in the main paper.

4. Grading Rubric (Total: 100 points)

Category	Points	Criteria
Problem & Contribution	10	Clear problem identification; well-articulated research gap; contribution is novel and significant relative to existing work
Related Work	10	Comprehensive coverage of relevant literature; proper positioning of contribution; critical analysis of prior work
Methodology & Design	15	Sound technical approach; clear threat model; appropriate design choices; well justified decisions
Implementation	20	Working, well-documented code; reproducible setup; appropriate use of tools/frameworks; addresses implementation challenges
Experimental Evaluation	25	Rigorous experimental design; appropriate datasets and metrics; fair comparison with baselines; thorough analysis of results; statistical significance; discussion of limitations
Paper Quality	10	Clear writing; logical structure; professional formatting; proper citations; effective use of figures/tables
Code & Reproducibility	10	Complete documentation; clear instructions; ability to reproduce key results; code organization

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Final project option 3: Network Simulation & Attack Analysis Lab Design

Objective: Design and implement a realistic network security lab environment that simulates a LAN/WAN architecture, executes cyber-attacks, and evaluates defense mechanisms through experimental analysis.

This project focuses on:

- Network architecture design
- Hands-on attack simulation
- Security monitoring and defense
- Data-driven evaluation

1. Project Overview

Students will create a virtual network lab that mimics a real-world environment (e.g., an enterprise or campus network), similar to the ones you have worked on in the cloud labs. then:

- Simulate network-based attacks
- Capture and analyze traffic
- Apply and evaluate defensive controls
- Document findings in an academic-style report

2. Suggested Tools

Students may use:

- **Mininet** (SDN-based lab environments)
- **GNS3 / EVE-NG** (realistic network emulation)
- **VirtualBox / VMware** (VM-based networks)
- **Docker** (containerized lab environments)

3. Attack Implementation (Choose ≥ 1)

Each project must simulate at least one attack from the following list or other attacks covered in class:

- DDoS (traffic flooding)
- Port scanning (e.g., reconnaissance)
- ARP spoofing
- DNS spoofing
- Man-in-the-Middle (MITM)
- Brute-force authentication attack
- Lateral movement inside LAN

4. Defense Mechanisms

Students must implement and evaluate at least one defense:

- Firewall configuration (e.g., rules, filtering)
- IDS/IPS (e.g., Snort, Suricata)
- Traffic monitoring (Wireshark, Zeek, tcpdump)
- Rate limiting / blocking strategies
- (Optional) AI/ML-based detection

5. Deliverables

The final deliverable is a lab manual in academic paper format (> 6 pages, single-column, double-spaced, 10-12pt font, Style: Clear, technical, reproducible) with the following components:

- **Introduction**
 - Describe the goal of your lab
 - Explain the network security problem you are studying
 - Provide motivation (e.g., why this attack/defense matters)
- **Network Design & Architecture**
 - Description of your network topology
LAN structure (clients, servers), WAN or external connection (if applicable)
 - Role of each component: Router, switch, firewall, IDS, hosts
 - **Must Include:**
 - Network topology diagram (clearly labeled)
 - IP addressing scheme
- **Tools & Environment Setup**
 - Tools used (e.g., Mininet, GNS3, VirtualBox, Docker)
 - System configuration: OS, VM/container setup
 - Network setup steps

This section should allow **someone else to reproduce your lab**
- **Attack Implementation**

Clearly describe **at least one attack scenario**

For each attack:

 - Objective of the attack
 - Tools/commands used
 - Step-by-step execution
 - Target and expected impact
- **Defense Mechanisms**

Describe and implement at least one defense, including:

 - Defense strategy (e.g., firewall, IDS, rate limiting)
 - Configuration details (rules, scripts, parameters)
 - Why this defense is appropriate
- **Experimental Setup**

Explain how experiments were conducted:

 - Test scenarios:
 1. Normal traffic
 2. Under attack
 3. With defense enabled
 - Tools used for measurement: Wireshark, tcpdump, Zeek, etc.
- **Results & Analysis**
 - Network Metrics: Latency, Throughput, Packet loss
 - Security Metrics: Detection rate, False positives / false negatives, Attack success rate
- **Comparative Analysis**
 - Before vs after attack
 - Before vs after defense
- Comparison across configurations

- Tables
- Graphs
- Screenshots (only if explained)
- **Discussion**
 - Key observations and insights
 - Strengths and weaknesses of your approach
 - Limitations of your lab setup
 - Real-world implications
- **Conclusion & Future Work**
 - Summarize findings
 - Suggest improvements
- **Appendix (Optional but Recommended)**
 - Configuration files
 - Scripts
 - Sample logs or packet captures
 - Additional screenshots
- Required Lab Artifacts (Submission Checklist)
- Students must submit:
 - Lab manual (PDF)
 - Network topology diagram
 - IP addressing table
 - Config files / scripts
 - (Optional) PCAP/log files

6. Grading Rubric (Total: 100 points)

Component	Weight
Network design & realism	20%
Attack implementation	15%
Defense mechanisms	15%
Experimental analysis	20%
Documentation quality	20%
Demo or recorded demo	10%

7. Academic Integrity

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